

Stating Measurable Learning Outcomes Using the Revised Bloom's Taxonomy

The Revised Bloom's Taxonomy (Anderson & Krathwohl, 2001) provides a structured framework for writing clear, specific, and measurable learning outcomes. It organizes cognitive learning into six hierarchical levels:

1. Remember
2. Understand
3. Apply
4. Analyze
5. Evaluate
6. Create

To make learning outcomes measurable, they must use observable action verbs and clearly describe what the learner will be able to do at the end of instruction.

1 Structure of a Measurable Learning Outcome

A well-written learning outcome typically includes:

Condition + Audience + Behavior + Degree (CABD Model)

Example Structure

By the end of the lesson, learners will be able to analyze case studies on occupational standards and identify at least four competency gaps correctly.

Where:

- **Condition** – Under what circumstances? (e.g., given a case study)
- **Audience** – Who? (e.g., learners, trainees)
- **Behavior** – Observable action (e.g., analyze, design, compare)
- **Degree** – Level of performance (e.g., with 80% accuracy)

2 Cognitive Levels with Action Verbs and Examples

2.1 Remember (Recall facts and basic concepts)

Action verbs: define, list, name, identify, recall, state

Example Outcomes:

- Define the term occupational standard.
- List the components of a competency framework.
- Identify five stakeholder groups involved in competency validation.

Measurable because: learners must demonstrate recall.

2.2 Understand (Explain ideas or concepts)

Action verbs: explain, describe, summarize, classify, discuss, interpret

Example Outcomes:

- Explain the relationship between competency elements and performance criteria.
- Summarize the process of stakeholder engagement in curriculum design.

- Classify different types of competencies in a given occupation.

Measurable because: can be assessed through written or oral explanation.

2.3 Apply (Use information in new situations)

Action verbs: apply, demonstrate, use, implement, calculate, solve

Example Outcomes:

- Apply Bloom's taxonomy to develop learning outcomes for a CBET module.
- Demonstrate how to formulate performance criteria from industry standards.
- Use a competency template to draft an occupational standard.

Measurable because: can be assessed through practical tasks or assignments.

2.4 Analyze (Draw connections among ideas)

Action verbs: analyze, differentiate, compare, contrast, examine, organize

Example Outcomes:

- Analyze the strengths and weaknesses of different curriculum design models.
- Compare competency-based education with traditional education models.
- Differentiate between core and functional competencies in a case study.

Measurable because: can be assessed through analytical reports or structured discussions.

2.5 Evaluate (Justify a decision or position)

Action verbs: evaluate, justify, assess, critique, defend, recommend

Example Outcomes:

- Evaluate the effectiveness of a stakeholder engagement plan.
- Critique a sample occupational standard using given criteria.
- Recommend improvements to a competency framework based on industry feedback.

Measurable because: assessed through evidence-based arguments.

2.6 Create (Produce new or original work)

Action verbs: design, develop, formulate, construct, create, propose

Example Outcomes:

- Design a competency-based assessment tool for a specific occupation.
- Develop measurable performance criteria for a selected job role.
- Construct a competency framework aligned with industry needs.

Measurable because: assessed through production of original work.

3 Characteristics of Measurable Learning Outcomes

Effective learning outcomes should be:

- **Specific** – Clear and focused
- **Observable** – Can be seen or measured
- **Measurable** – Includes criteria for success
- **Learner-centered** – Focus on learner performance
- **Aligned** – Linked to assessment and instruction

4 Avoid Non-Measurable Verbs

Avoid vague verbs such as:

- Understand
- Know
- Appreciate
- Be familiar with
- Learn

Instead of:

Learners will understand competency frameworks.

Write:

Learners will explain the key components of a competency framework with at least three examples.

5 Sample Outcomes for a CBET Module

Module: Occupational Standards and Competency Framework

By the end of the module, learners will be able to:

1. Define key terms related to occupational standards. (*Remember*)
2. Explain the components of competency standards. (*Understand*)
3. Apply stakeholder engagement principles to competency identification. (*Apply*)
4. Analyze industry needs to determine required competencies. (*Analyze*)
5. Evaluate the relevance of existing occupational standards. (*Evaluate*)
6. Design a competency-based occupational standard for a selected occupation. (*Create*)

6 Key Takeaway

To state measurable learning outcomes using Revised Bloom's Taxonomy:

- Choose the appropriate cognitive level.
- Use observable action verbs.
- Clearly describe the expected performance.
- Include measurable criteria where possible.
- Align outcomes with assessment methods.

Well-written learning outcomes improve clarity, guide instruction, and ensure meaningful assessment in competency-based education and training (CBET).